

Kazi Sher Ahmed

Mechatronics Researcher

Karachi, Pakistan

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RESEARCH INTERESTS	Active magnetic bearings (AMB), electric machine design, ultra-high-speed motors & spindles, control systems, rotordynamics	
EDUCATION	Doctoral Studies – Sabanci University, Turkiye Mechatronics Engineering (coursework completed) - Research Topic: Optimization and Realization of AMB Spindle for Chatter Suppression - Advisor: Dr. Bekir Bediz - Relevant Coursework: Machine Learning, Metal Cutting Mechanics, Tribology, Scaling in Energy Systems - Status: Paused in Fall 2023 following a family bereavement	Sep. 2021 – July 2023
	MS – Sabanci University, Turkiye Mechatronics Engineering - Thesis: Design of Active Magnetic Bearing Spindles for Micro-Milling Applications - Advisor: Dr. Bekir Bediz - Relevant Coursework: Real-Time Systems, Integrated Sensors, Mechanical Vibrations, Topology Optimization, Finite Element (FE) Method, Machine Tool Engineering	Sep. 2018 – July 2021
	BS – GIK Institute, Pakistan Mechanical Engineering - Senior Year Project: Development & Finite Element Modeling of a Rotor-Bearing Test Rig - Award: Second Best Senior Year Project	Aug. 2011 – June 2015
JOURNAL PAPERS	K S Ahmed & B Bediz, 2026, ‘ Design framework and manufacturing of an active magnetic bearing spindle for micro-milling applications ’, <i>The International Journal of Advanced Manufacturing Technology</i> K S Ahmed , G Kiziltas, & B Bediz, 2025, ‘ Integrated design optimization of a motorized active magnetic bearing spindle for micro-milling applications ’, <i>Journal of Mechanical Science and Technology</i> , vol. 39, pp. 2015–2027 K S Ahmed & S M Ahmad, 2019, ‘ VibronRotor, an open-source rotordynamic code: Development and benchmarking ’, <i>Measurement</i> , vol. 131, pp. 546–558	
CONFERENCE PAPERS	K S Ahmed & B Bediz, 2022, ‘ Design of an active magnetic bearing spindle for micro-milling applications ’, <i>19th International Conference on Machine Design and Production</i> , Cappadocia, Turkiye, Aug. 31–Sep. 3 A Rehman, K S Ahmed , F A Umrani, B Munir, A Mehboob, S M Ahmad, & Z Kazmi, 2015, ‘ Finite element modelling of a generic rotor-bearing system and experimental validation ’, <i>ASME Dynamic Systems and Control Conference</i> , Ohio, USA, Oct. 28–30	
RESEARCH PROJECTS	Design Framework & Manufacturing of AMB Spindle Graduate Researcher [Papers: IJAMT UMTIK] - Proposed an iterative framework to design and manufacture micro-milling AMB spindles, aiming to consolidate fragmented design knowledge - Demonstrated the framework by prototyping a 5-DOF spindle designed for 110,000 rpm with permanent-magnet-biased homopolar AMBs and air turbine drive. Corrected analytical model agreed with 3D electromagnetic finite element (FE) model within 1.17% in AMB radial force - Built the prototype through CNC milling, cylindrical grinding, and wire EDM. Validated structural FE models through experimental modal testing	Sabanci University 2019 – 2023

	Integrated Design Optimization of Motorized AMB Spindle Graduate Researcher [JMST Paper]	Sabanci University 2022 – 2023
	- Co-wrote a successful grant proposal (TUBITAK Grant: 123M121) - Developed a multidisciplinary optimization framework integrating AMB and motor performance, conventionally treated separately, for a 5-DOF miniaturized spindle - Employed analytical models covering electromagnetics, losses, mechanics, and rotordynamics - Constructed a 3D finite-element model to: (i) analyze and reduce AMB-motor electromagnetic coupling, (ii) numerically correct the AMB analytical model - Generated Pareto-optimal designs trading motor constant against spindle mass, validated against single objective optimization at extreme points (maximum deviation 4.1%)	
	Open-Source Rotordynamic Software Laboratory Engineer [Measurement Paper] [Code]	GIK Institute 2017 – 2018
	- Developed and benchmarked VibronRotor, an open-source finite-element rotordynamic code compatible with MATLAB and the free scientific programming language GNU Octave - Implemented key analyses: Campbell diagram, critical speed map, mode shapes, imbalance response, orbit plots, and instability threshold analysis - Instrumented a rotor test rig with eddy current probes for experimental modal analysis for code validation	
INDEPENDENT PROJECT	Open-Source DC Motor Control Test Rig [Code]	2026
	- Built a low-cost rig for DC motor characterization and embedded control, with STM32 Nucleo-F446RE, BTS7960 H-bridge power stage, ACS712 current sensing, and quadrature encoder feedback - Identified motor parameters from locked-rotor, steady-state, and coast-down tests. Validated the model against the measured position response to a voltage step ($R^2 = 0.998$) - Implemented PI speed control, pole-placement and LQR position control, and Luenberger/Kalman state estimation. Compared four position-control configurations on matched 0.20 rad steps - Tested an aggressive LQR tuning with encoder-derived and Kalman-estimated speed feedback. Both exhibited persistent oscillation at the duty limits, so observer-based pole placement was adopted for this rig	
SKILLS	<i>Methods:</i> Control system design & embedded implementation, electromagnetics, FE analysis, rotordynamics, design optimization, experimental modal analysis <i>Programming & numerical computing:</i> MATLAB, Python, C/C++, Mathematica <i>CAD & simulation:</i> SolidWorks, COMSOL, ANSYS, MATLAB/Simulink <i>Experimental & data acquisition:</i> Test rig instrumentation, LabVIEW	
PEER REVIEW SERVICE	Journal of the Brazilian Society of Mechanical Sciences and Engineering, Springer Journal of Mechanical Science and Technology, Springer	
WORK AND TEACHING EXPERIENCE	Independent Researcher Completed and published two journal papers from my graduate school research. Built an open-source motor control rig. Reviewed manuscripts for two Springer journals	Jan. 2024 – Present
	Graduate Researcher & Teaching Assistant, Sabanci University	Sep. 2018 – Dec. 2023
	- Research outcome: 2 journal articles (IJAMT 2026, JMST 2025), 1 conference paper (UMTIK 2022) - Weekly recitations, grading, and office hours for courses: Control Systems, Mechanical Vibrations, Linear Algebra, Statics, Dynamics, Mechanical Systems, Discrete Mathematics, Calculus - Mentored undergraduate students on the development of an active magnetic damper	
	Laboratory Engineer, GIK Institute	Sep. 2016 – Aug. 2018
	- Research outcome: 1 journal article (Measurement 2019), open-source rotordynamic code - Designed and conducted lab experiments in Mechanical Vibrations and Fluid Mechanics - Teaching assistant course assignments: Fluid Mechanics, Mechanical Vibrations - Served as Problem-Based Learning Committee Coordinator	

- Project Engineer, GIK Institute** Sep. 2015 – Aug. 2016
- Designed and conducted computational fluid dynamics analysis of a micro-hydro Kaplan turbine for a Khyber-Pakhtunkhwa Govt. Project
 - Delivered a seminar talk on ‘Indigenous R&D of Micro-Hydro Power Plants’, CIIT Lahore (2016)
- Summer Intern, K-Electric Combined Cycle Power Plant** June 2014 – July 2014
- Completed on-site trainings on plant operations and delivered technical reports
- Research Intern, MARC, Koc University** July 2013 – Aug. 2013
- Designed and conducted stress analysis of a dynamically-loaded spring for linear compressors
- HammerHead Arc** (Shell Eco-marathon team)
- Team manager (2014–15), Team member (2013–14)
 - Led the team and represented Pakistan in Shell Eco-marathon 2015, Philippines
 - Designed, fabricated, and tested an ultra energy-efficient electric vehicle
- Managing Editor, Science Aurora Magazine** Aug. 2012 – May 2013
- Researched content, wrote articles, and managed the project

AWARDS

- Graduate studies tuition waiver & stipend support at Sabanci University, Turkiye (2018–2023)
- Second position in Senior Year Project Competition at GIK Institute (2015)
- Represented Pakistan in Shell Eco-marathon Asia, Manila, Philippines (2015)
- Best Design Award in National Shell Eco-marathon Off-track Event, Pakistan (2015)
- Sixth position in Battery Electric Category, 2014 Shell Eco-marathon Asia, Manila. Project also won Best Ecological Footprint Award at National Shell Eco-marathon event (2014)

TEST SCORES TOEFL: 106/120 (2021)

REFERENCES **Dr. Bekir Bediz**
Associate Professor, Sabanci University, Turkiye
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Dr. Gullu Kiziltas Sendur
Professor, Sabanci University, Turkiye
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